```
# 1. Update system
sudo apt update

# 2. Install required tools
sudo apt install -y gnupg software-properties-common curl

# 3. Add HashiCorp GPG key
curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/hashicorp-archive-keyring.gpg

# 4. Add repository
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee
/etc/apt/sources.list.d/hashicorp.list

# 5. Update again
sudo apt update

# 6. Install Vagrant
sudo apt install vagrant
—----------------------------
vagrant --version
—----------------------------
Mkdir jenkins-vagrant-lab
cd jenkins-vagrant-lab
vagrant init ubuntu/jammy64


_____________________
sudo apt install virtualbox
vboxmanage --version
_____________________
vagrant init
vagrant up

Vargrantfile content:

Vagrant.configure("2") do |config|

  config.vm.box = "ubuntu/jammy64"

  config.vm.provision "shell", inline: <<-SHELL
    sudo apt update
    sudo apt install -y openjdk-17-jdk openssh-server
```

```
sudo systemctl start ssh
sudo systemctl enable ssh


    mkdir -p /home/vagrant/jenkins
    chmod 777 /home/vagrant/jenkins
  SHELL

  config.vm.define "master" do |master|
    master.vm.hostname = "master"
    master.vm.network "private_network", ip: "192.168.56.11"
  end

  config.vm.define "agent1" do |a|
    a.vm.hostname = "agent1"
    a.vm.network "private_network", ip: "192.168.56.12"
  end

end
```

—----------
If ownership is not given do this

```
 sudo chown -R user:mca /home/mca/jenkins-vagrant-lab
mca@administrator:~$ chmod -R u+rwX /home/mca/jenkins-vagrant-lab
```

—----------

```
vagrant up
vagrant status
```

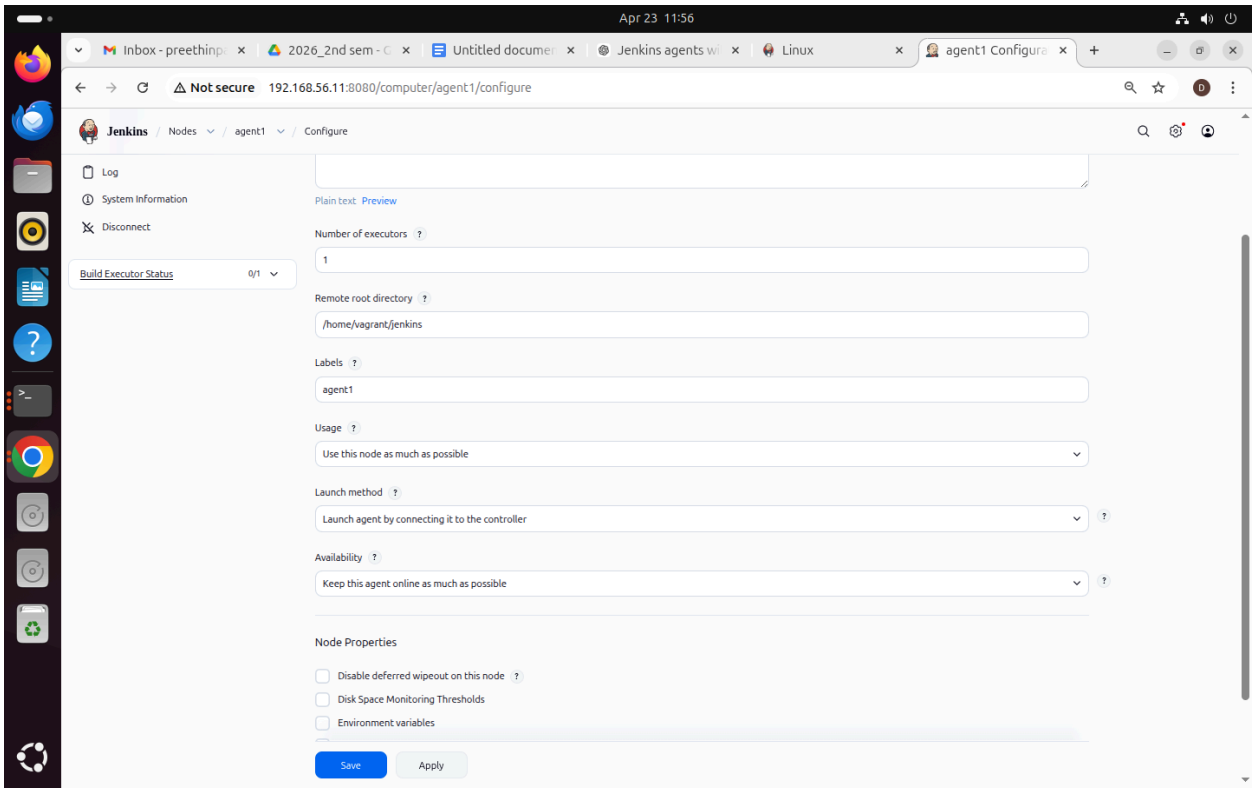
—----------
vagrant ssh master–on new terminal
—----------
vagrant ssh agent1–on another terminal
—----------
On master: make sure java is installed→ sudo apt install openjdk-21-jre -y

Install jenkins–follow the steps from [jenkin.io](jenkin.io)

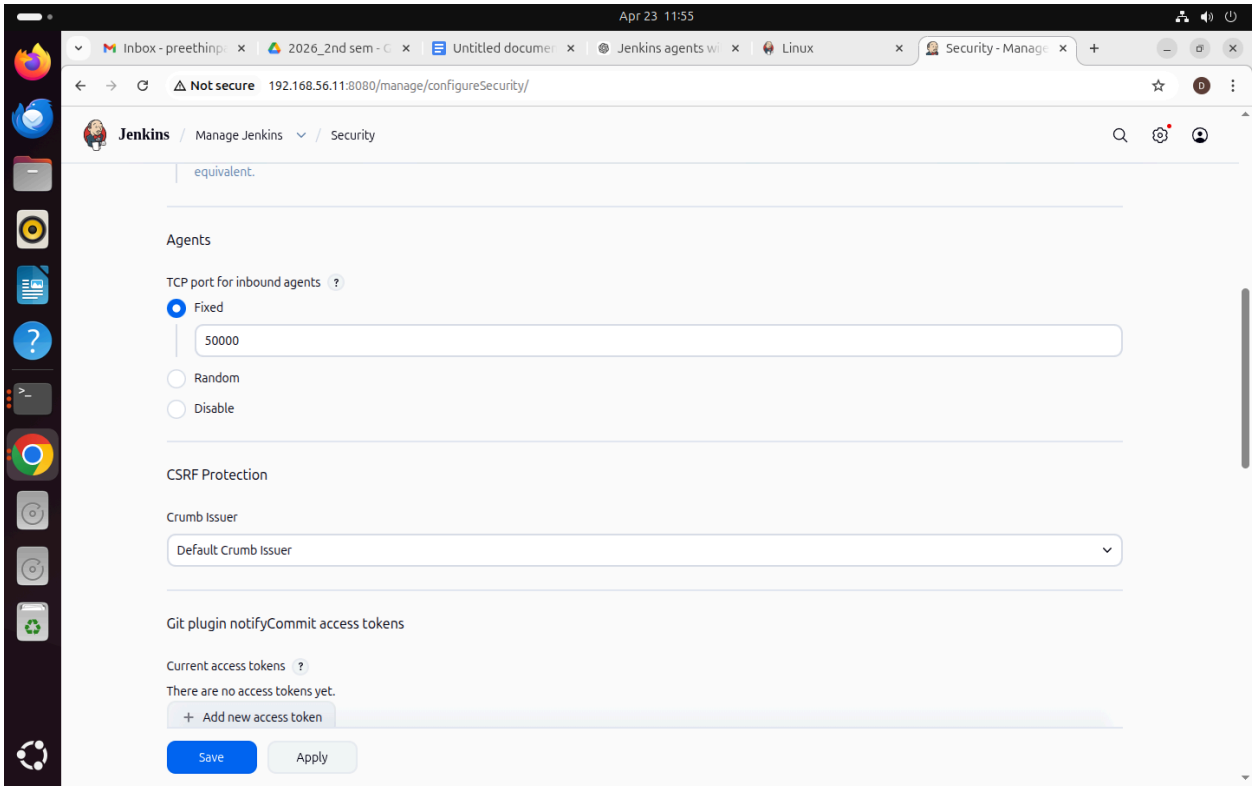
```
sudo systemctl start jenkins
sudo systemctl enable jenkins
```

And then launch jenkins on hppt://192.168.56.11:8080

In jenkins create new node




Under security do this change




On agent terminal

wget [http://192.168.56.11:8080/jnlpJars/agent.jar](http://192.168.56.11:8080/jnlpJars/agent.jar)

On master terminal
Copy paste the key on the jenkins agent1
Some thing like this

```
curl -sO http://192.168.56.11:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://192.168.56.11:8080/ -secret
a12ae4c377d6b03a36202594bffbce228421940aec7036858e8ab6d2a69f1c25 -name agent1
-webSocket -workDir "/home/vagrant/jenkins"
```

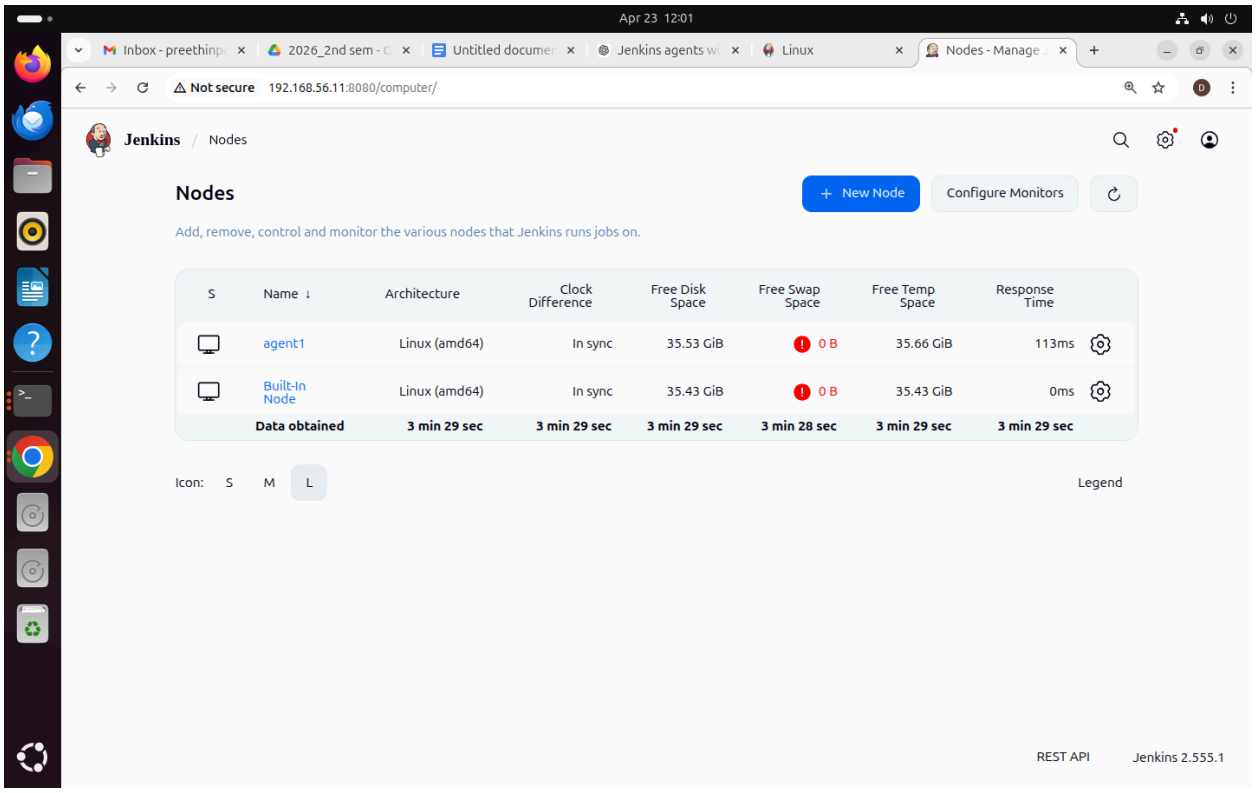
---

To test the setup add groovy script in the jenkins instance (test-agent under the pipelinescript section)

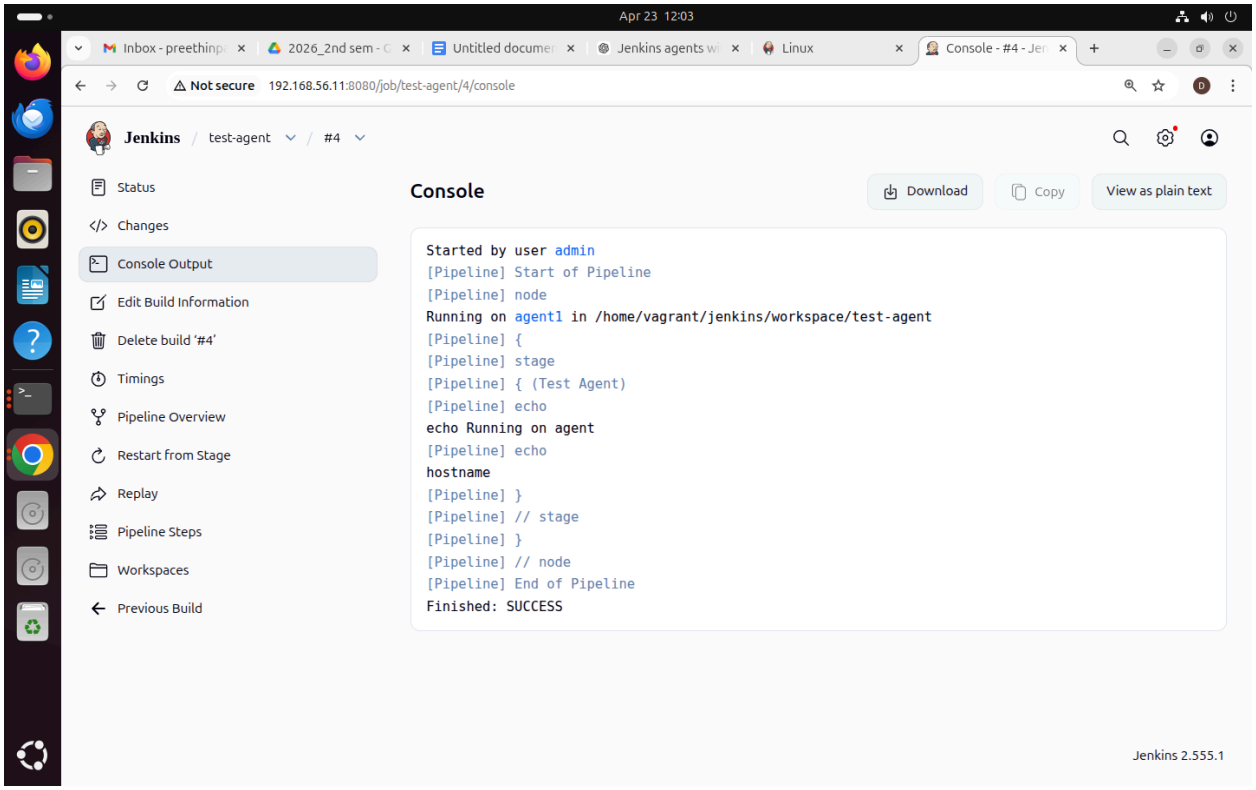
```
pipeline {
    agent { label 'agent1' }
    stages {
        stage('Test Agent') {
            steps {
                sh 'echo Running on agent'
```

```
            sh 'hostname'
        }
    }
  }
}
```

Then agent becomes online




Final console output


Apr 23 12:03
Inbox - preethinp
2026_2nd sem - C
Untitled docume
Jenkins agents wi
Linux
Console - #4 - Jen
Not secure
192.168.56.11:8080/job/test-agent/4/console
Jenkins
test-agent
#4
Status
Changes
Console Output
Edit Build Information
Delete build '#4'
Timings
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on agent1 in /home/vagrant/jenkins/workspace/test-agent
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Test Agent)
[Pipeline] echo
echo Running on agent
[Pipeline] echo
hostname
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.555.1